

SIEMENS

MAGNETOM

MR

Installation and Startup

MAGNETOM Option

UPS Powerware 9130G-3000T-XLEU

Prisma, Prisma fit
Aera, Skyra, Avanto fit, Skyra fit
Biograph mMR
Avanto, Espree, Verio, Essenza
Symphony, A Tim System; Trio, A Tim System
Harmony, Symphony, Sonta, 7T, Trio
Allegra, Numaris 3.5 systems, MAGNETOM C! Mod.2

© Siemens, 2008

All documents may only be used for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

Print No.: MR00-000.814.42.05.02
Replaces: MR00-000.814.42.04.02

English
Doc. Gen. Date: 08.13

Copyright

“© Siemens, 2008” refers to the copyright of a Siemens entity such as Siemens Aktiengesellschaft - Germany, Siemens Shenzhen Magnetic Resonance Ltd. - China, Siemens Shanghai Medical Equipment Ltd. - China, Siemens Medical Solutions USA Inc. - USA, Siemens Healthcare Diagnostics Inc. - USA and/or Siemens Healthcare Diagnostics Products GmbH - Germany.

Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed. Please contact your local Siemens office to order current version or refer to our website <http://www.healthcare.siemens.com>.

Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The installation and service of equipment described herein requires superior understanding of our equipment and may only be performed by qualified personnel who are specially trained for such installation and/or service.

1	General Information	4
	Safety Information	4
	General	5
	Required tools and auxiliary tools	5
	Time required	5
	Scope of delivery	5
	UPS system for the Host PC	6
2	Installation	9
	Installation	9
	Overview	9
	Overview Biograph mMR	10
	Wiring Diagram	10
	Installation and Cabling	14
	Start-up	17
	Concluding Tests	17
3	Changes to the previous version	18
4	List of Hazard IDs	19

Safety Information

For your safety and the safety of third parties as well as for protecting the MR system and its components from damage, the appropriate safety information is included in register “General Safety Notes” and “MR-Specific Safety Information” in the “Technical Documents”.

Prior to starting work, read the specific information in the respective manual:

- General safety information for service personnel.
- Limitations with respect to ferromagnetic materials.
- Safety information pertaining to working with the magnet at field.
- Information regarding working with heavy loads.
- Protective conductor test before switching on the system and for subsequent repairs.

General

The documentation describes the installation of the Powerware PW 9130G 3000T-XLEU®.

Required tools and auxiliary tools

- Tool kit
- Wall bracket installation (optional):
 - 8 mm drill
 - Power drill

Time required

The time required for the option is approx. 2 hours.

Scope of delivery

NOTE

Check the parts using the material list supplied with the delivery volume.



Fig. 1: UPS Powerware PW 9130



Fig. 2: Distribution box



Fig. 3: Cable set

UPS system for the Host PC

NOTE

If an older UPS and distribution box (with another material no. as below) was used, it must always be replaced, together with the distribution box, by the UPS Powerware PW 9130G 3000T-XLEU®.

Distribution box:

- Aera, Skyra, Tim systems, Avanto^{fit}, Skyra^{fit}, Prisma and Prisma^{fit}::

Material no.: 81 14 717

- all other systems:

Material no.: 104 33 233

NOTE

When installing an MRSC console, UPS Powerware PW 9130G 3000T-XLEU® does not support the MRSC PC. You must install a standard PC UPS or additional UPS Powerware PW 9130G 3000T-XLEU® (without a cable set) to supply the MRSC PC or a Leonardo workplace computer.

The UPS Powerware PW 9130G 3000T-XLEU® is used for the MRC Host computer.

Function

Using a UPS (uninterrupted power supply) ensures that the system computer is protected against all types of interferences.

In most cases, local voltage supplies are sufficient for satisfactory operation.

However, modern processor-controlled configurations require highly reliable sources. The Powerware PW 9130G 3000T-XLEU® meets these requirements. The

Powerware PW 9130G 3000T-XLEU ® is the standard UPS for the **MRC Host PC and Imager PC**.

NOTE

The PW 9130G 3000T-XLEU® including the Siemens cable set supports the Host PC and the Imager PC.

For other options, e.g. MRSC, procure an additional UPS system, preferably a PW 9130G 3000T-XLEU®.

Technical data

Technical data of Powerware 9130G 3000T-XLEU®	
Weight in kg	36
Dimensions of housing H x W x D in mm	325 ¹ x 214 x 412
Output in kVA	3.0
Backup time in min.	at max. load 5 min. with built-in batteries
Temperature	+0°C to +40°C
Relative humidity	5% - 90% noncondensing
Acoustic noise	<50 dBA ²

1. 346 mm with feet
2. at 1.0 meter distance

Order information**NOTE**

The item numbers and descriptions of the PM (Price Book) apply.

Order information		Order-related details
Type	Material no.:	Consulting the price book
Powerware 9130G 3000T-XLEU® (including wall bracket)	144 13 662	
Battery module PW 9130G-3000T-EBM	144 13 663	
Siemens cable set (is included with the connection box)	88 57 828 (Aera, Skyra, Tim sys- tems, Avanto ^{fit} , Skyra ^{fit} , Prisma, Prisma ^{fit} and Biograph mMR) 144 13 664 (all other systems)	

NOTE

The distribution box for Powerware 9130G 3000T-XLEU® is part of the delivery volume for the Siemens cable set.

Installation

NOTICE

The Powerware 9130G-3000T-XLEU® Output Voltage is configured to 208 V for delivery.

Failure to observe the following safety information may result in equipment damage.

⇒ Adapt the output voltage of the Powerware to 230 V.

Overview

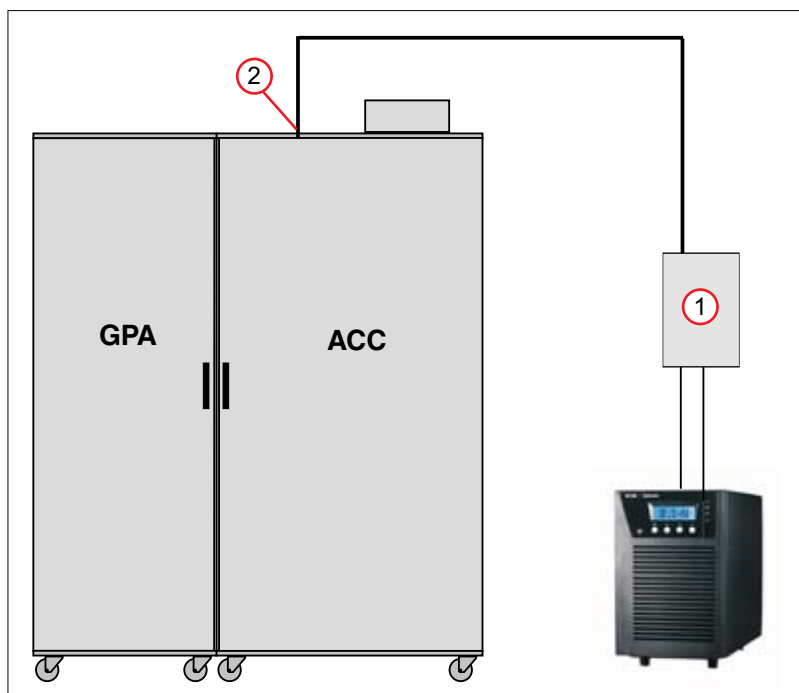


Fig. 4: UPS configuration for the host computer

Pos. 1 Distribution box

Pos. 2 To ACC/EPC X112; CCA X103

Overview Biograph mMR

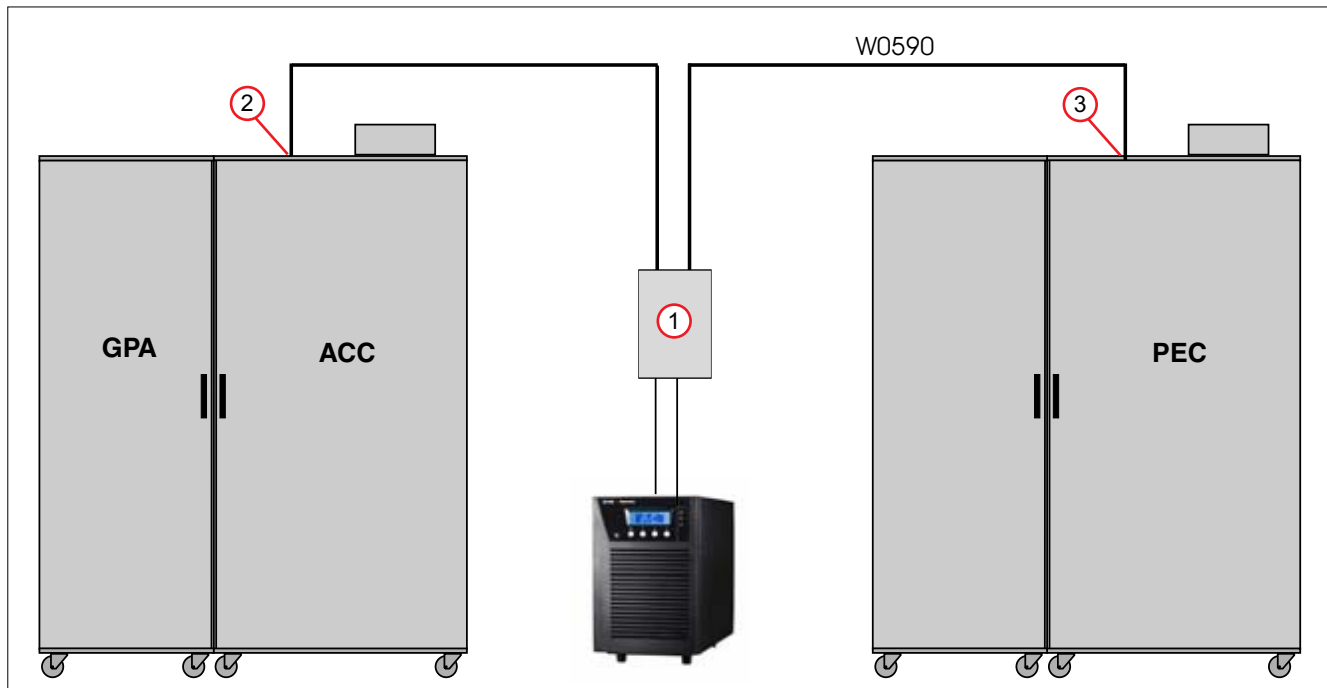


Fig. 5: UPS configuration with PEC (Biograph mMR)

- Pos. 1 Distribution box
 Pos. 2 To ACC X112
 Pos. 3 To PEC X38

Wiring Diagram

NOTE

If a small UPS (for the host PC) is supplying the system, pressing the EPO button interrupts the power supply between the “Distribution Box” and the CCA (Symphony) or ACC (TIM-Systems) or EPC (Aera/Skyra/Avanto^{fit}/Skyra^{fit}/Prisma/Prisma^{fit}) cabinet.

Only for Biograph mMR:

The PEC cabinet is equipped with a separate small built-in UPS system.

This UPS system has to be switched off by activating the EPO button as well.

NOTE

Only for older Verio systems:

If no EPO switch is used, the contact of the distribution box must be bridged in the location, refer to (1/ Fig. 6 / p. 11).

For all other (non Tim) systems, refer to (Fig. 8 / p. 13).

For Aera, Skyra, Tim systems, Avanto^{fit}, Skyra^{fit}, Prisma and Prisma^{fit}:

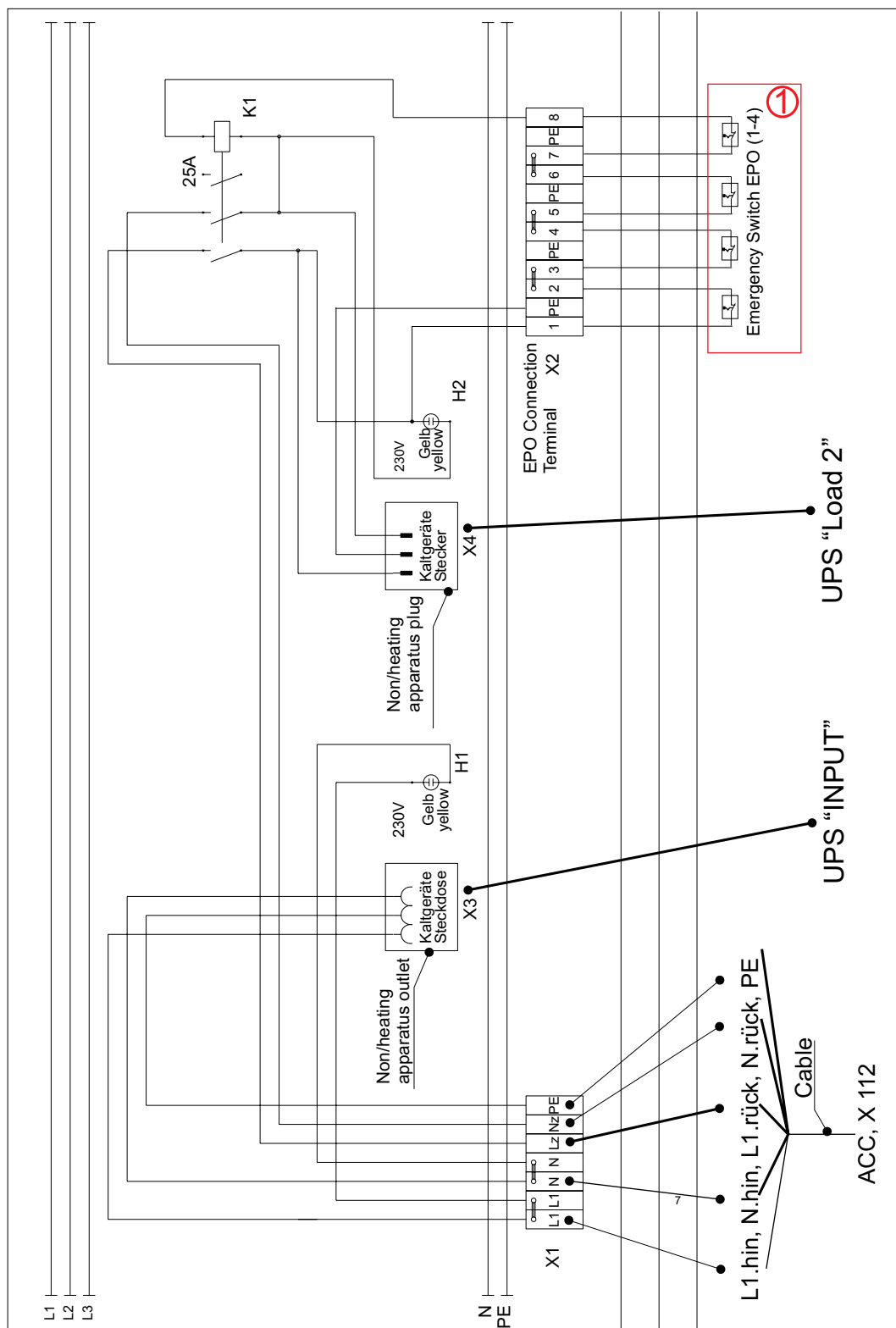


Fig. 6: Distribution box (material no. 8114717)

Pos. 1 A total of four EPO switches can be connected to the terminal block X2. Jumper the respective EPO terminal block with a bridge if no EPO is connected to that terminal

For Biograph mMR

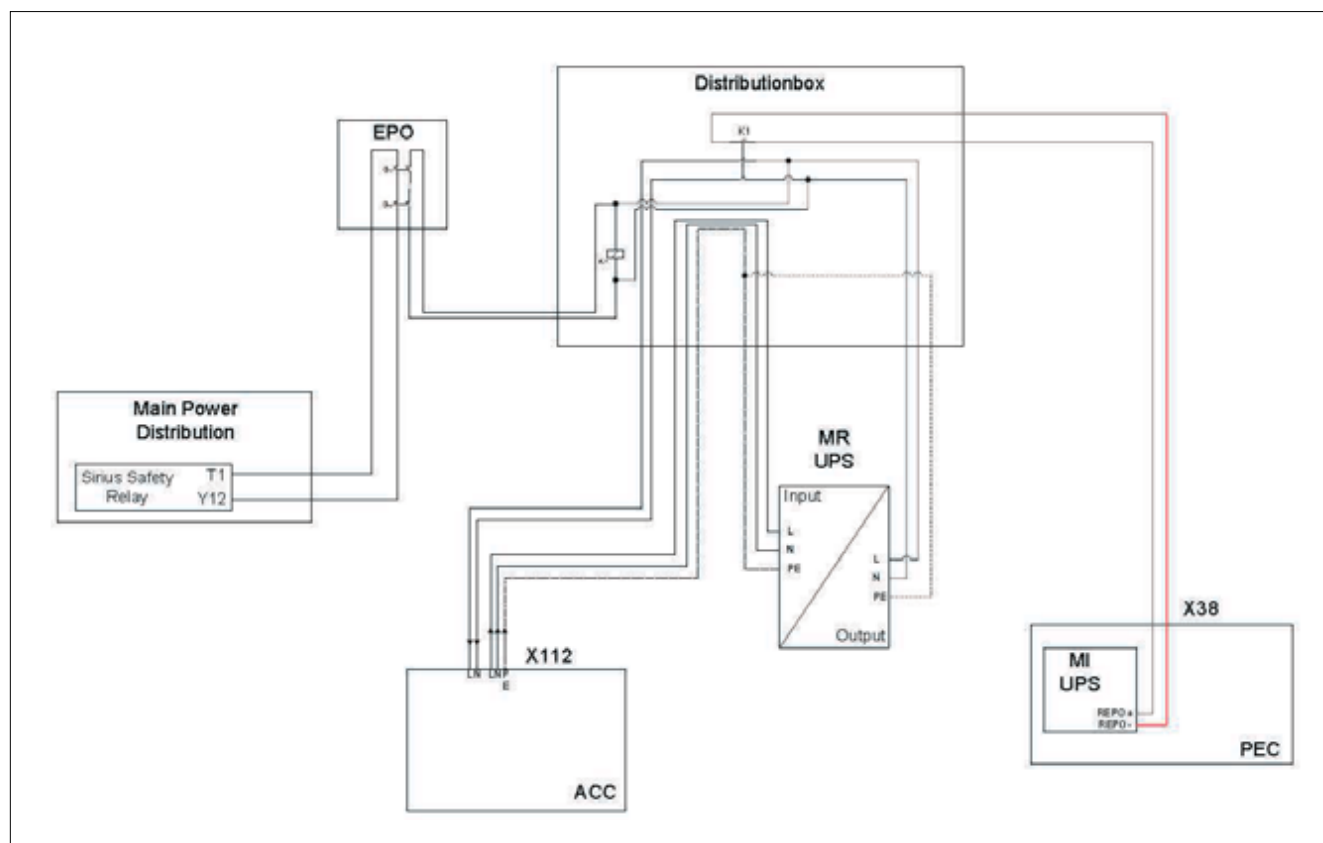


Fig. 7: Connecting the PEC UPS to the distribution box

For all other Systems

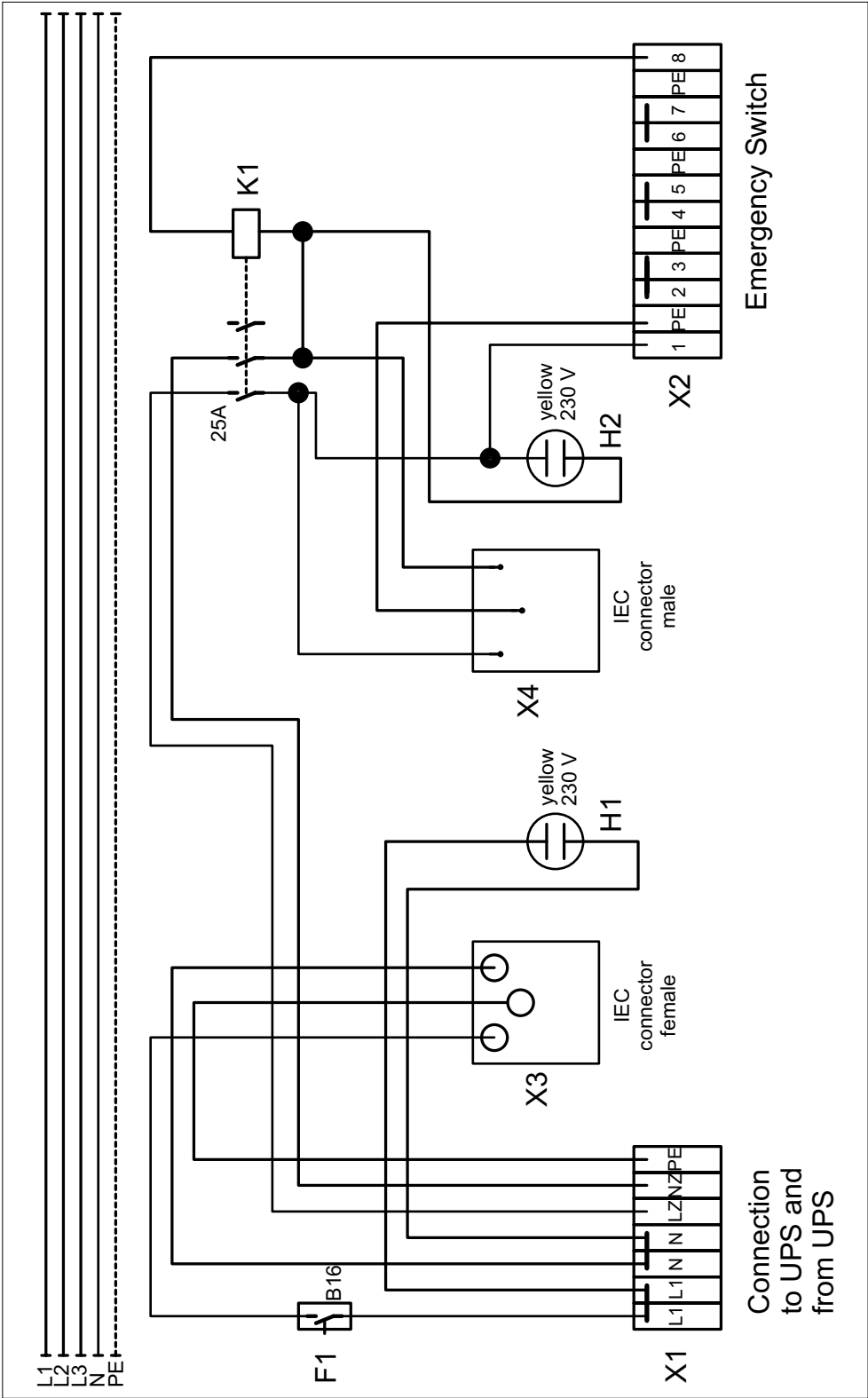


Fig. 8: Distribution box (material no. 10433233)

Installation and Cabling

NOTE

For additional information, please refer to the documentation for the components of UPS Powerware 9130G 3000T-XLEU® supplied with the system.

- Deenergize the system.
- Connect the line voltage to terminal X1 , refer to

Cable identifications	Designation of distribution box Terminal X1
L1.hin	L1
N.hin	N
L1.rück	Lz
N.rück	Nz
PE	PE

Only for Biograph mMR:

- Connect cable W0590 to terminal X3, refer to

Aera, Skyra, Tim systems, Biograph mMR, Avanto^{fit}, Skyra^{fit}, Prisma and Prisma^{fit}::

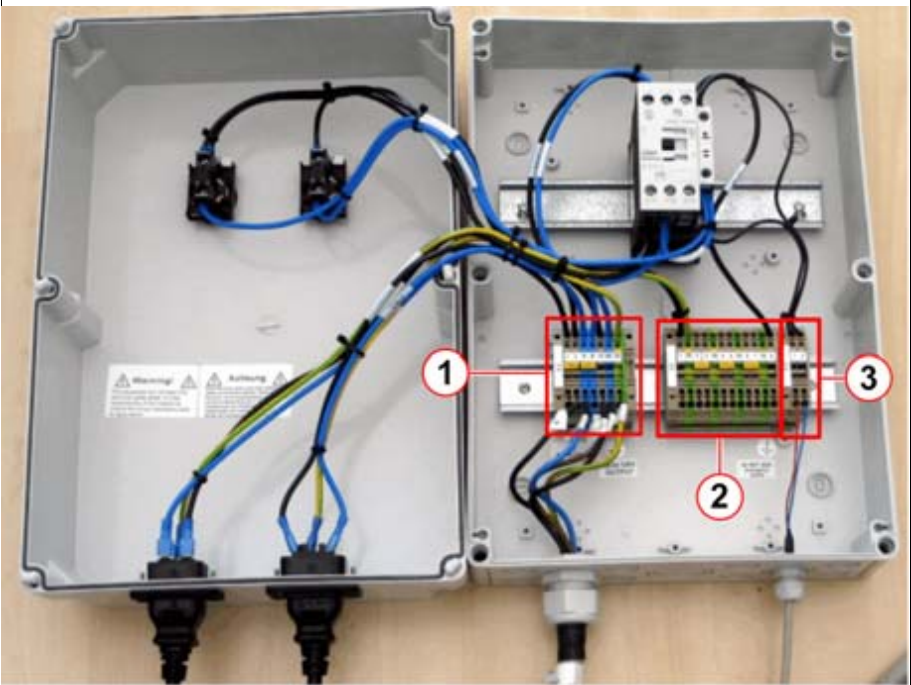


Fig. 9: Distribution box (material no. 8114717)

Pos. 1 X1 (cable connection to ACC/EPC X112)

Pos. 2 X2 (connection terminals for EPO switches)

Pos. 3 X3 (connection terminal for PEC UPS)

For all other systems:

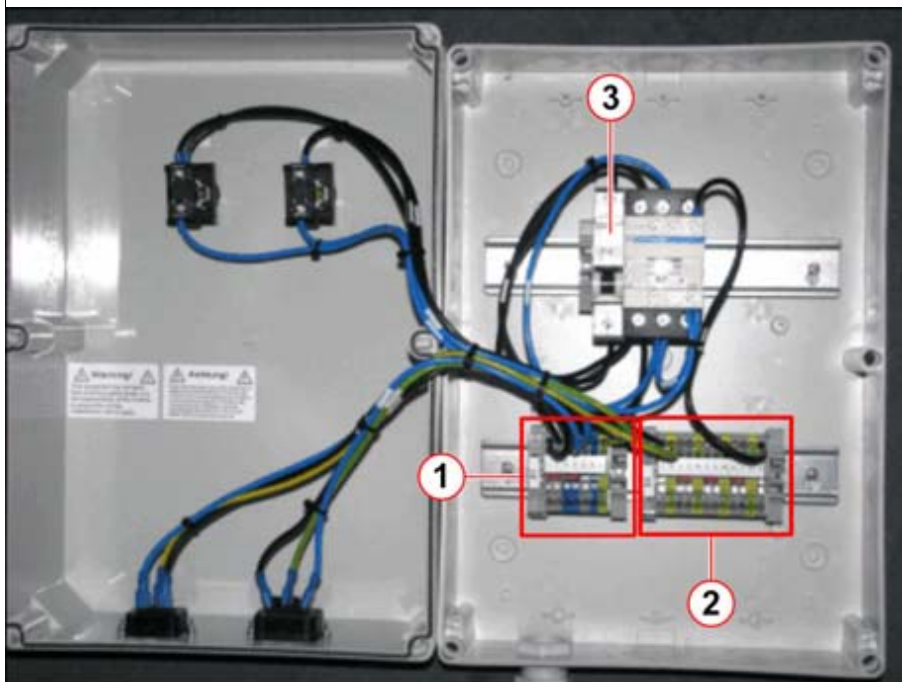


Fig. 10: Distribution box (material no. 10433233)

- Pos. 1 X1 (cable connection to CCA X103)
 Pos. 2 X2 (connection terminals for EPO switches)
 Pos. 3 F1 16A



Fig. 11: Distribution box

- Pos. 1 To ACC / EPC X112; CCA X103
 Pos. 2 X3 to UPS INPUT
 Pos. 3 X4 to UPS LOAD2
 Pos. 4 To PEC X38 (optional for Biograph mMR)

- Connect the distribution box to Pow-erware 9130:
 - Distribution box X3 - UPS INPUT
 - Distribution box X4 - UPS LOAD2
- Connect the distribution box to the EPC / ACC / CCA cabinet:

Distribution box (1) to

- EPC X112 (Aera, Skyra, Avanto^{fit}, Skyra^{fit}, Prisma and Prisma^{fit})
- or
- ACC X112 (Tim systems and Bio-graph mMR)
- or
- CCA X103 (all other systems)

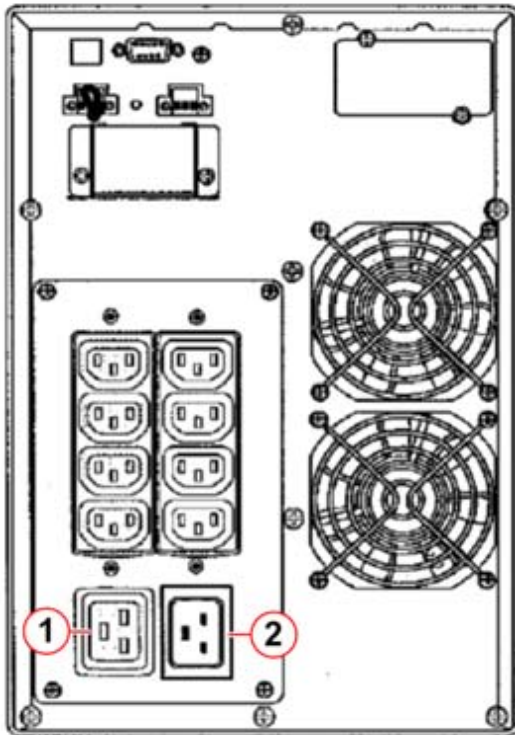
Only for Biograph mMR:

- Connect the distribution box to the PEC cabinet:

Distribution box (4) to

- PEC X38 (W0590)

Connecting the supplied cables:	
Distribution box	UPS
X3	"INPUT"
X4	"LOAD2"



- Switch the system on.
- Switch on the Powerware UPS.

Fig. 12: UPS Powerware 9130i 3000T-XLEU (rear side)

- Pos. 1 Output LOAD2 to distribution box X4
Pos. 2 INPUT to distribution box X3

Start-up

NOTICE

The Powerware 9130G-3000T-XLEU® Output Voltage is configured to 208 V for delivery.

Failure to observe the following safety information may result in equipment damage.

⇒ Adjust the output voltage of the Powerware to 230 V.

Adjust the output voltage to 230 V, refer to the Powerware User Guide supplied with the system. The output voltage adjustment is described in the user settings.

- Adjust the output voltage from 208 V to **230 V**.

Concluding Tests

A distinction is made between systems with an EPO and those with a main system switch (only with Verio systems).

Systems with EPO

NOTE

The UPS system and the MR system must be switched off when the EPO switch (Emergency Power Off) is switched off.

- ⇒ The UPS system must be switched off. The UPS system must stop supplying the MR system with voltage once the EPO switch is activated.

Systems with main system switch (only with Verio systems)

NOTE

The UPS system and the MR system must be switched off when the main system switch is switched off.

- ⇒ The UPS system must be switched off. The UPS system must stop supplying the MR system with voltage when the main system switch is turned off.

Version 02

- Aera and Skyra added

Version 03

- Biograph mMR added
- Editorial changes.

Version 04

- The following systems were added: Avanto^{fit} and Skyra^{fit}

Version 05

- The following systems were added: Prisma and Prisma^{fit}
- Minor changes

There are no Hazard IDs in this document.

